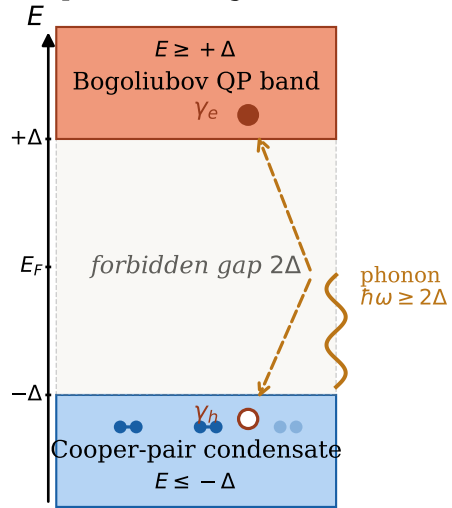
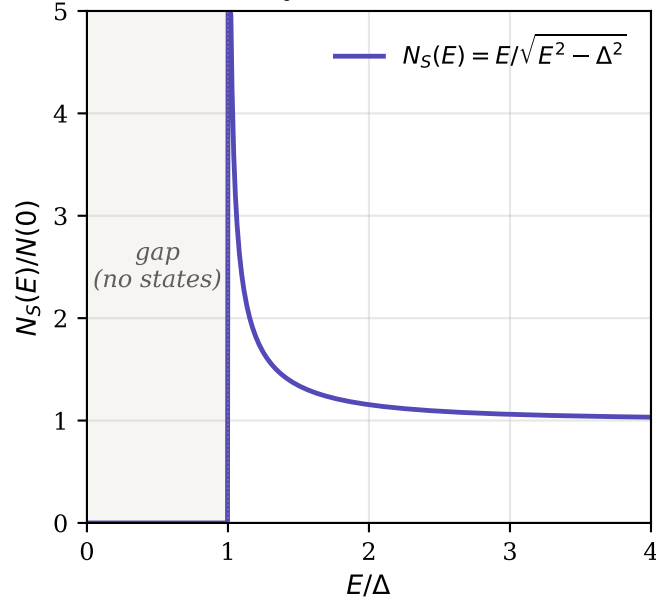


(a) pair breaking



(b) BCS density of states

(c) population = $N_S \times f$ 